

VanRakshak: AI-Enabled Drone Patrolling System for Forest Authorities to Detect Poaching, Illegal Logging, and Encroachment in Real Time

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VANRAKSHAK

AI-Enabled Drone Patrolling System for Forest Authorities to Detect Poaching, Illegal Logging, and Encroachment in Real Time

AI-Powered Autonomous Wildlife Intelligence Platform

MSME Idea Hackathon 6.0
Theme: Other Frontier Technologies

Saveetha Engineering College
Department of CSE

Building India's Indigenous Wildlife Intelligence Platform

To build India's leading indigenous AI-powered wildlife intelligence platform that enables autonomous forest surveillance, protects biodiversity, and assists authorities in preventing poaching, illegal logging, forest fires, and encroachment through real-time edge AI.

15 km²

Coverage per sortie

<3 sec

Alert latency

95%+

Detection accuracy

THE PROBLEM

Forests Are Vast. Surveillance Isn't.

India's forests span vast, often inaccessible terrain. Manual patrols, static CCTV, and occasional drone flights leave critical gaps — poaching, illegal logging, encroachment, and early-stage fires go undetected until it's too late.



Limited Coverage

Vast, remote forest terrain is impossible to patrol continuously on foot.



Personnel Shortage

Forest departments are chronically understaffed for the area they guard.



Delayed Detection

Incidents are discovered hours or days after they occur.



Poor Night Visibility

Most poaching and logging activity happens after dark.



No Intelligent Analysis

Raw footage with no automated threat detection or prioritization.



High Patrol Risk

Ground teams face direct physical danger confronting armed poachers.

Existing Solutions Fall Short

Manual Patrol

Low coverage · no night vision · no AI · high human risk

Static CCTV

Fixed coverage · no autonomy · needs manual review

Commercial Drones

Records video only — no onboard intelligence or real-time alerts

The gap: no existing system autonomously detects, classifies, and alerts on forest threats in real time — that's exactly what VanRakshak delivers.

VanRakshak

AI-Powered Autonomous Wildlife Intelligence Platform

VanRakshak combines autonomous drones, multi-sensor fusion, edge computing, and intelligent threat analytics to monitor forests in real time — turning raw footage into actionable intelligence for forest authorities.

-  **Autonomous Drone Patrol**
-  **Edge AI Inference**
-  **GPS-Tagged Alerts**
-  **Multi-Sensor Fusion**
-  **Real-Time Threat Detection**
-  **Mission Planning & Dashboard**

AT A GLANCE

Sector
Other Frontier Technologies

Prototype Compute
NVIDIA Jetson Orin Nano

Core Sensors
RGB · Thermal · LiDAR · GPS · IMU

Primary Users
Forest Departments, National Parks

Ask
₹15 Lakhs (MSME Hackathon)

From Sensor Data to Field Response

1

Patrol mission assigned to the drone fleet

2

Drone autonomously flies predefined forest routes

3

RGB, thermal & LiDAR sensors capture live data

4

Edge AI performs onboard inference

5

Suspicious activity detected & confidence scored

6

GPS-tagged alert with image/video evidence generated

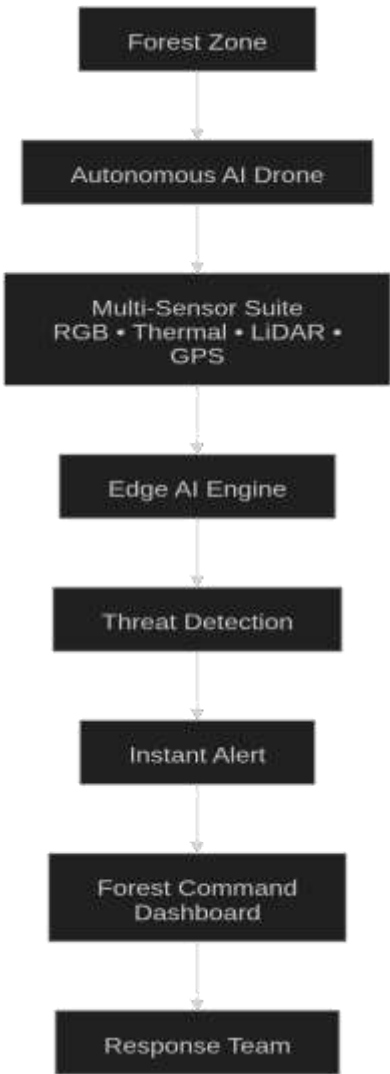
7

Alert reaches the Forest Command Dashboard

8

Response team is dispatched

Product Architecture



Forest Zone → Drone

Autonomous drone launches to patrol a defined forest zone.

Multi-Sensor Suite

RGB, thermal, LiDAR & GPS capture a full picture of the terrain.

Edge AI Engine

Onboard inference detects threats without needing constant connectivity.

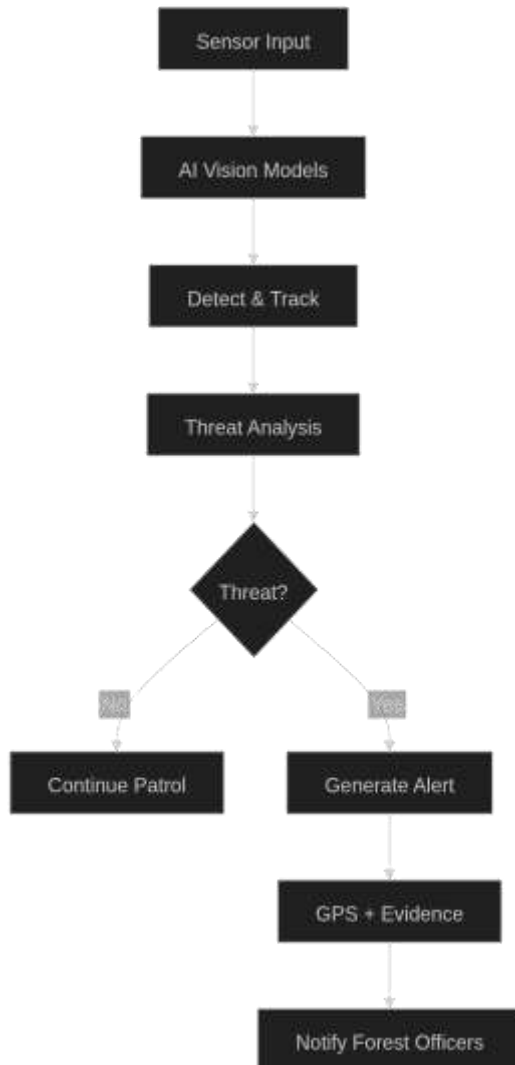
Alert → Dashboard

Instant, GPS-tagged alerts reach the Forest Command Dashboard.

Response Team

Officers are dispatched with precise location and visual evidence.

AI Pipeline



AI Vision Models

YOLO & RT-DETR object detection tuned for forest environments.

Detect & Track

Multi-object tracking follows movement across frames.

Threat Analysis

Behavioral and contextual signals classify the activity type.

Confidence Scoring

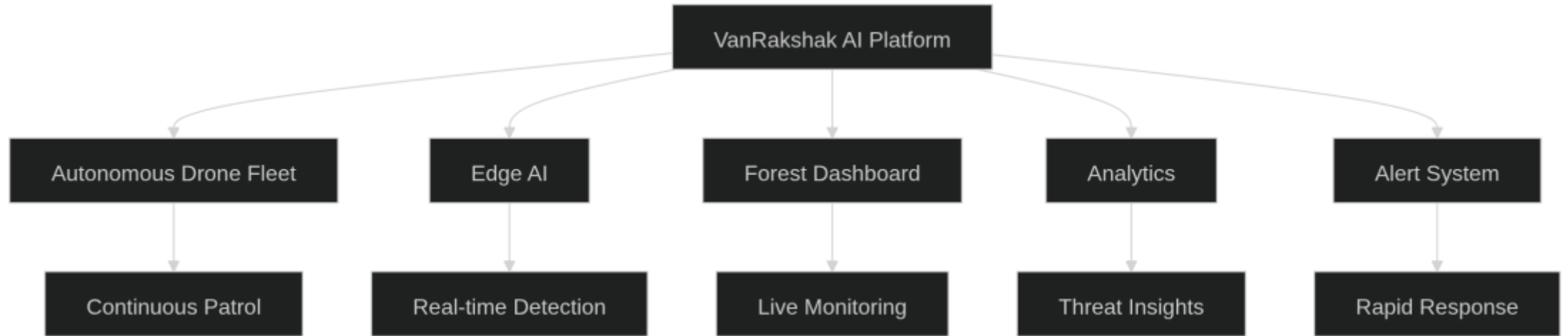
Every detection is scored to minimize false alarms (<5%).

Alert Generation

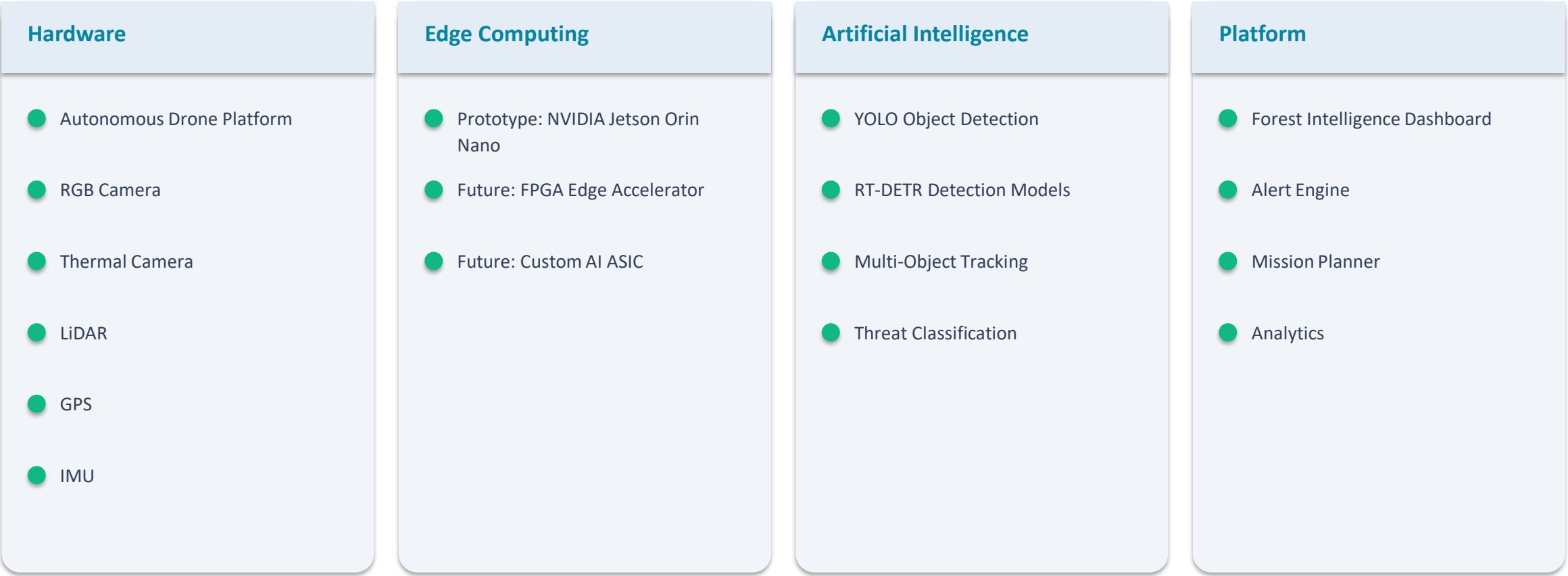
High-confidence threats trigger GPS-tagged alerts with evidence.

Product Ecosystem

VanRakshak is a platform, not just a drone — five connected pillars work together.



Technology Stack



Why VanRakshak Wins

Capability	Manual Patrol	CCTV	Commercial Drone	VanRakshak
Coverage	Low	Fixed	Medium	High
Autonomy	None	None	Partial	Full
Night Vision	No	Limited	Rare	Yes
AI Detection	No	No	No	Yes
Edge Computing	No	No	No	Yes
Real-time Alerts	No	No	No	Yes
Continuous Monitoring	No	Partial	No	Yes

Who We Serve, and Why It Matters

1

Target Customers

State Forest Departments, Wildlife Protection Authorities, National Parks; expanding to NGOs, eco-tourism, mining, railways & disaster management.

2

Pain Points

Vast unmonitored terrain, staff shortages, delayed incident detection, and dangerous manual patrols in remote regions.

3

Value Proposition

Continuous autonomous monitoring with real-time AI threat detection — faster response, lower risk, wider coverage.

4

Revenue Opportunity

Rising government investment in AI, drones & smart conservation technology fuels demand for scalable indigenous platforms.

Multiple Revenue Streams, Built for Scale

PRIMARY REVENUE

- ₹ Drone Hardware Sales
- ₹ Drone-as-a-Service (DaaS)

RECURRING REVENUE

- 🔄 Annual Maintenance Contracts
- 🔄 AI Analytics Subscription
- 🔄 Training Programs
- 🔄 Government Contracts

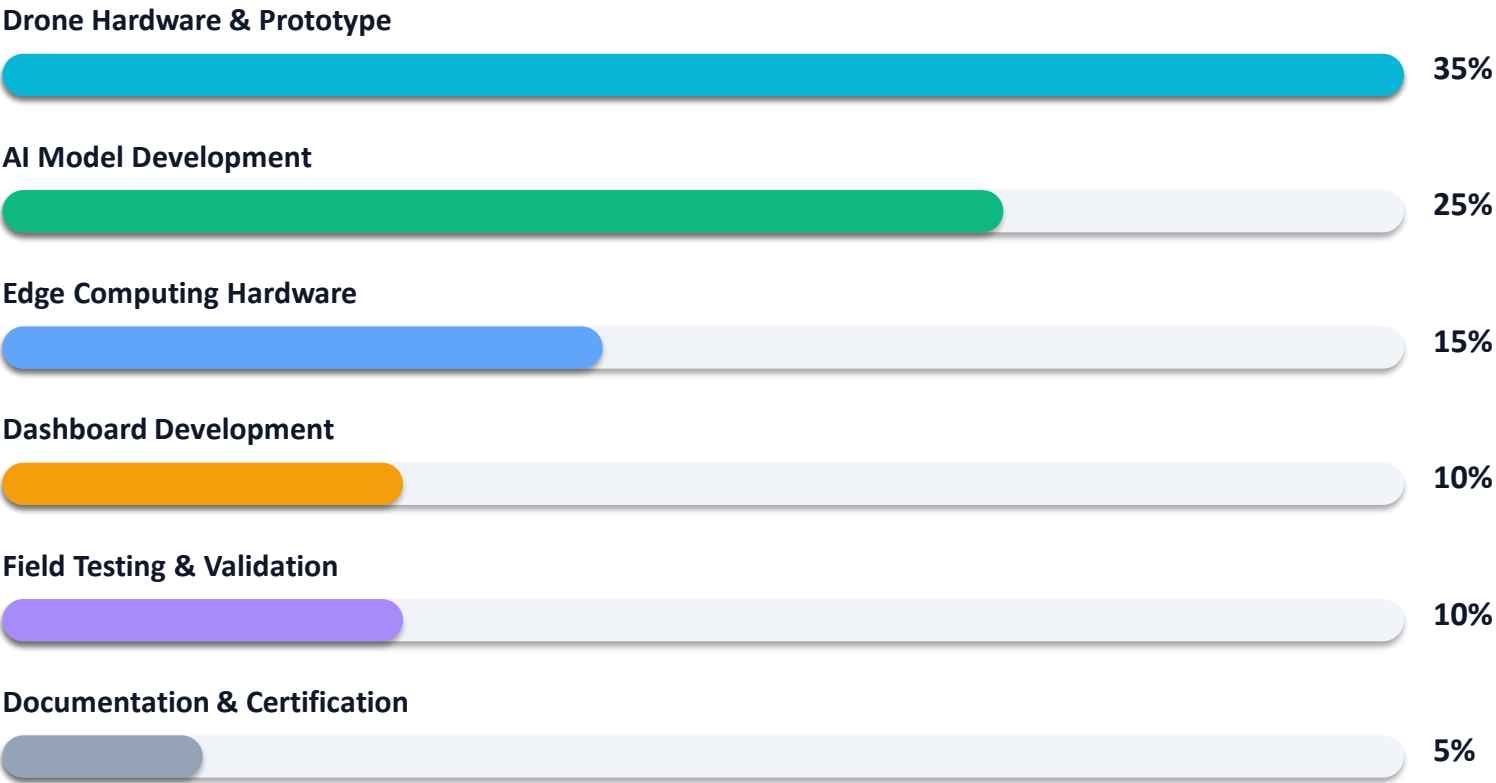
From Prototype to National Deployment



FUNDING ASK

Seeking ₹15 Lakhs

MSME Idea Hackathon 6.0 — Proposed Allocation



TOTAL ASK

₹15,00,000

Funds the drone prototype, edge AI stack, dashboard build, and field validation with partner forest departments through pilot deployment.

Targets That Prove the Platform Works

>95%

Detection Accuracy

<3 sec

Alert Latency

>15 km²

Coverage per Sortie

>45 min

Flight Time

<5%

False Alarm Rate

>60%

Patrol Cost Reduction

From a Drone Solution to a National Wildlife Monitoring Network



VanRakshak

“From Surveillance to Intelligence.”

Protecting India's forests through autonomous AI.